

Multiple Choice Quiz

Time and Clocks in Distributed Systems

Instructions: For each of the following 10 questions, select the **best** answer from the four options provided (A, B, C, or D). Mark your choice clearly. Each question tests conceptual understanding from the lecture on physical time, NTP, TrueTime, happens-before, Lamport clocks, and Vector clocks. There is exactly one correct answer per question.

Question 1. Why is the banking thought experiment used to illustrate the problem of physical time?

- A) It proves that physical clocks are perfectly synchronized using GPS signals in banking systems worldwide.
 - B) To show how network latency always causes transaction failures in banks regardless of clock synchronization.
 - C) To illustrate that banking systems require atomic clocks on every server to prevent overdrafts globally.
 - D) It demonstrates the difficulty of ensuring a debit on one server happens before a credit on another server across the world. It highlights that without a global "now," ordering transactions across distributed servers is not guaranteed by physical time alone.
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Question 2. Define the terms "skew" and "drift" in the context of physical clocks.

- A) Skew is the statistical variance between multiple clock readings, and drift is the name of the synchronization protocol used (e.g., NTP drift compensation).
 - B) Skew means how much a clock lags behind real UTC time, and drift is the network propagation delay experienced during NTP synchronization.
 - C) Skew refers to the specific difference in time between two clocks at any given moment. Drift is the rate at which that skew increases over time, typically caused by imperfections in quartz crystal oscillators.
 - D) Skew is the average time difference observed over a 24-hour period, while drift refers to sudden manual adjustments made by system administrators.
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Question 3. What is the hierarchy used by the Network Time Protocol (NTP), and what resides at the top?

- A) NTP uses a flat peer-to-peer model where every server is considered equal; Stratum 1 servers act as the primary reference clocks synced directly to UTC.
 - B) The hierarchy starts with Stratum 0 as ordinary client machines; Stratum 1 servers sit below them and sync upward to GPS sources.
 - C) It employs a strict tree hierarchy with Stratum 1 devices (atomic clocks) at the very top, followed by Stratum 2 servers that clients connect to.
 - D) NTP uses a hierarchical stratum model to organize time servers. At the top of this hierarchy is Stratum 0, which consists of high-precision timekeeping devices like atomic clocks and GPS clocks.
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Question 4. How does Google Spanner's TrueTime mechanism handle the inherent uncertainty of physical clocks?

- A) It relies exclusively on vector clocks and logical timestamps, completely avoiding any dependence on uncertain physical clock values.
 - B) Spanner achieves perfect clock synchronization across all nodes using dedicated atomic clocks and therefore reports exact timestamps with zero uncertainty.
 - C) TrueTime acknowledges clock error by returning time as an uncertainty interval $[T, T+\epsilon]$. Transactions are required to wait out this uncertainty interval before committing to ensure external consistency.
 - D) Transactions commit immediately using the local clock; if skew is later detected, conflicting transactions are automatically rolled back.
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Question 5. What are the two conditions that define the "happens-before" relation ($\rightarrow$)?

- A) Events are related by happens-before if they execute on different processes and their physical timestamps differ by less than the measured network round-trip time.
 - B) The relation holds when both events access the same shared data item and one process acquires a distributed lock before the other performs its update.
 - C) Happens-before is defined solely by comparing physical timestamps from a globally synchronized clock; the earlier timestamp precedes the later one.
 - D) The first condition is that two events occur on the same process, with one naturally preceding the other. The second condition is that one event is the sending of a message and the other is the receipt of that message by another process.
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Question 6. Explain the update rule for a Lamport scalar clock when a process receives a message.

- A) The receiving process simply increments its local clock by one upon message arrival and ignores the timestamp value carried by the message.
 - B) It adds the received clock value to the local counter and then adds an extra increment equal to the total number of processes in the system.
 - C) The local clock is set directly to the received clock value if it is strictly greater; otherwise the clock remains unchanged (no increment occurs).
 - D) When a message is received, the process compares its own local counter (C) with the counter attached to the incoming message (received C). It updates its local counter to be the maximum of those two values, plus one ($C := \max(C, \text{received } C) + 1$).
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Question 7. What is the primary "catch" or limitation when using Lamport scalar clocks for ordering?

- A) Lamport clocks only produce correct orderings when every process already possesses a perfectly synchronized physical clock (e.g., via GPS).
 - B) They require a centralized coordinator process to collect and merge clock values from all participants, introducing a single point of failure.
 - C) The limitation is that Lamport clocks cannot distinguish between causal relationships and concurrency. While they provide a total ordering, a smaller clock value does not guarantee that one event actually caused the other; they might have happened independently.
 - D) The primary drawback is excessive network overhead: every message must carry a very large timestamp whose size grows with the number of processes.
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Question 8. How does the structure of a Vector Clock differ from a Lamport scalar clock?

- A) A Vector Clock still uses only one integer like Lamport but augments it with a floating-point drift-rate value obtained from the local quartz oscillator.
 - B) Vector clocks store timestamps as floating-point numbers to represent fractional logical time, replacing the simple integer counter used by Lamport clocks.
 - C) While a Lamport clock uses a single integer counter for the entire process, a Vector Clock maintains an array (vector) of counters. Each entry in the vector corresponds to one specific process in the distributed system, tracking what that process knows about every other process.
 - D) Vector clocks are maintained by a single centralized server that tracks every process's logical time, whereas Lamport clocks are updated locally by each process.
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Question 9. According to the comparison rules, when are two events considered concurrent in a Vector Clock system?

- A)** Events are concurrent if neither vector is strictly "less than" the other. Specifically, if one vector has some components that are larger than the other's, but also has some components that are smaller, the events have no causal relationship.
 - B)** Concurrency exists when the sum of all vector components is identical for both events, regardless of the distribution across individual processes.
 - C)** Two events are declared concurrent precisely when every component of their vector clocks contains exactly the same integer value.
 - D)** Events are considered concurrent only when they were generated by the same process and their vector entries differ by exactly one in a single component.
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Question 10. In what types of systems is the ability to detect concurrency particularly important?

- A)** Concurrency detection matters most in centralized mainframe batch-processing systems to improve CPU utilization of long-running jobs.
 - B)** The capability is primarily needed in real-time multiplayer video games so that all players observe events in identical physical-time sequence.
 - C)** Detecting concurrency is critical for distributed databases and collaborative editing systems. In these environments, the system must identify when multiple users have made independent changes so it can resolve potential conflicts.
 - D)** It is especially vital in single-threaded embedded controllers where minimizing power consumption and interrupt latency is the top priority.
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Answer Key

This answer key is provided for instructors or self-assessment. The correct option letter is shown for each question after randomization of choice order. The full correct answer text is included for reference.

Q1. Correct Answer: D

It demonstrates the difficulty of ensuring a debit on one server happens before a credit on another server across the world. It highlights that without a global "now," ordering transactions across distributed servers is not guaranteed by physical time alone.

Q2. Correct Answer: C

Skew refers to the specific difference in time between two clocks at any given moment. Drift is the rate at which that skew increases over time, typically caused by imperfections in quartz crystal oscillators.

Q3. Correct Answer: D

NTP uses a hierarchical stratum model to organize time servers. At the top of this hierarchy is Stratum 0, which consists of high-precision timekeeping devices like atomic clocks and GPS clocks.

Q4. Correct Answer: C

TrueTime acknowledges clock error by returning time as an uncertainty interval $[T, T+\epsilon]$. Transactions are required to wait out this uncertainty interval before committing to ensure external consistency.

Q5. Correct Answer: D

The first condition is that two events occur on the same process, with one naturally preceding the other. The second condition is that one event is the sending of a message and the other is the receipt of that message by another process.

Q6. Correct Answer: D

When a message is received, the process compares its own local counter (C) with the counter attached to the incoming message (received C). It updates its local counter to be the maximum of those two values, plus one ($C := \max(C, \text{received } C) + 1$).

Q7. Correct Answer: C

The limitation is that Lamport clocks cannot distinguish between causal relationships and concurrency. While they provide a total ordering, a smaller clock value does not guarantee that one event actually caused the other; they might have happened independently.

Q8. Correct Answer: C

While a Lamport clock uses a single integer counter for the entire process, a Vector Clock maintains an array (vector) of counters. Each entry in the vector corresponds to one specific process in the distributed system, tracking what that process knows about every other process.

Q9. Correct Answer: A

Events are concurrent if neither vector is strictly "less than" the other. Specifically, if one vector has some components that are larger than the other's, but also has some components that are smaller, the events have no causal relationship.

Q10. Correct Answer: C

Detecting concurrency is critical for distributed databases and collaborative editing systems. In these environments, the system must identify when multiple users have made independent changes so it can resolve potential conflicts.